CS 170 HW12

1.

Adversary wants to maximize regret, so $c_1^{(t)}=1$, and $c_i^{(t)}=0$ for all ~~t~~ $i>1$

The total cost is the sum of all t_i, which is 1, per timestep, so its T.

$\ell = \frac{1}{T}(T-0)=1$

b) ~~Let~~ For any day i, you will choose some expert j with 100% probability. Set $c_j=1$, and $c_{i\neq j}=0$ for all $i\neq j$. The ~~total cost~~ Regret is $\frac{1}{T}(T-0)=1$, since $\sum_{i=0}^{T} 1 - \sum_{i=0}^{T} 0 = T$.

c) An equal distribution minimizes regret. This is because for some p_k with the lowest probability, the casino assigns a cost of 0, and it assigns a cost of 1 to every other expert. Hence, the expected regret is $(1-p_k)$ where p_k is the smallest probability. The largest p_k is $\frac{1}{n}$, ~~hence~~ which is a uniform distribution, where expected regret is $1-\frac{1}{n}$. Hence, it must be randomized, since otherwise $p_k=0$, expected regret $=1$. The algorithm must also be adaptive, to choose expert k with a probability higher than p_k over time.

2 W is the set of all experts. If the majority of experts are wrong, we make a mistake. We also remove atleast half of the experts because they were wrong. Hence, we can make atmost $\log n$ mistakes before we are left with only the expert who is always right, so we make no more mistakes.

b) Using the same reasoning, W is the set of all experts and atleast half of the experts are removed per mistake. Since W has a size of atleast k, since k experts are always right, $k \leq \frac{n}{2^m}$, $2^m \leq \frac{n}{k}$, m, which is the number of mistakes, $m \leq \log(\frac{n}{k}) = \log n - \log k$. So the mistakes are bounded by $\log(n) - \log(k)$

c) During each step where $m \leq K$, we only listen to experts who have made m mistakes, if we got it wrong half of our experts who have made m mistakes are no longer in W_m, or the experts with only m mistakes. So L_m decreases by a factor of atleast 2 per mistake. If $L_m = 0$, then $m \leftarrow m+1$, until $m > K$. Hence, we take $\log(n)+1$ steps per m and we make at most $k+1$ m's, so our bound is $\log(n+1)(k+1)$. Hence our total mistakes are bounded by $\log(n+1)(k+1)$.